Launching Your Career in High-Performance Computing

In-Person Experience for Students at RMACC Institutions – Hosted by CU Boulder

A one-week experience to provide hands-on experience and direct access to subject-matter experts in high-performance computing (HPC), enabling participants to begin careers in HPC and related fields.

How to use this booklet

This booklet is designed for the in-person experience of the RMACC student cohort program, taking place at the University of Colorado Boulder and emphasizing technologies used on-site by their Research Computing team. This version has been prepared for the January 2026 session. The exercises and other materials might be of value separate from the experience, but they are necessarily incomplete without it. Material is offered as-is. Reader discretion advised.

Workshop policies

Ethical guidelines apply. Do not plagiarize. Treat people respectfully. Cite the NSF grant in any materials meant for distribution, following the pattern in "Acknowledgements" near the end of this document. Consider reading the USENIX System Administrator's Code of Ethics.

Program

Computing Platforms for Research

There are a lot of ways one could group research computing resources into discrete categories. The one we'll use is as follows:

1. Personal computing: Modern laptops, desktops, tablets, and phones all (to differing degrees) offer immense computing power and can be used for research purposes. In this category, we might also include "closet clusters" managed and administered directly by the user(s).
2. Cloud computing: Most modern web services are hosted "in the cloud", which generally refers to renting the usage of hardware and software managed by a third-party commercial provider. Examples include Microsoft Azure, Amazon Web Services (AWS), and Google Cloud Platform. These significantly ease administrative overhead for users but can be cost-prohibitive for high-utilization systems.
3. On-premises (on-prem) high-performance computing (HPC) systems: An organization (such as a university, department, corporation, or laboratory) owns physical hardware placed in a data

center and provides most or all administration directly. The primary HPC systems at CU Boulder are Alpine and Blanca.

This workshop focuses predominantly on category 3, though categories 1 and 2 follow most of the same principles.

We will begin by verifying sign-in to OnDemand and to XDMoD.

Hardware

Key ideas in this section: Moving forward and backward in time. Process of going from getting hardware to scheduling a job on that hardware. CPU. GPU. Data centers: sites, cabinets, racks, chassis, nodes, switches. Includes a hardware demo – see and touch some servers and components.

Exercise 1: Choose your hardware

1(a): Quote review

For this homework please see the mock quote in the appendix. Instructors will act as representatives of other teams as we conduct the exercise.

You have been assigned to complete a purchase of high-memory nodes. You have contacted vendor Acme, Inc., who has provided a quote for you to evaluate. **The quote contains a minimum of 5 fixes that need to be made prior to it being an acceptable quote for purchase. Please identify these issues and tell us how you would correct them.** You are free to use any of the resources we have provided to you. As a hint, it's good to ask yourself and your peers two questions:

1. Does the quote contain *all* the items we *need*?
2. Does the quote *exclude* any items that we *do not need*?

Please work in groups of four and be prepared to ask questions.

1(b): Proposal for hardware

You have now learned how to read and understand a quote. Next, we will be providing a recommendation for a hardware purchase.

You are assigned to spend $560,000 on a hardware order for a research computing HPC cluster. **Using the specifications in the table in the appendix, please prepare an order recommendation.** You do not need to create a quote. Your target audience is a reviewer, such as a director or other person with financial approval authority, who wants to be sure we're getting the best we can for the money we have. Focus on the high-level specifications for node components (e.g. CPU, storage, InfiniBand).

For this assignment, please write down the specifications you chose and explain why you made these choices. You will verbally present these details to us and at the end of the exercise you will present to the group. Designate one spokesperson for your group, with the other members ready to answer follow-up questions as needed.

Provisioning

Key ideas: Operating systems. Health checks. Imaging. Stateless and diskless.

Exercise 2: Provision the hardware with Warewulf

You have recently completed an order for CPU nodes for your HPC center. The order was accepted by all participants, produced, shipped, and delivered. You have worked with your team to rack, stack, cable, and validate in the data center. You are now prepared to begin work on releasing the node into production for your users.

In this exercise, you will develop and configure an OS and associated software using stateless and diskless imaging. The tool, in this instance, will be the software package Warewulf and its associated utility wwctl.

We will pair up, thus forming approximately 20 groups, each assigned to a single node and a single image. We have 32 nodes in reserve for these projects. Node and image names are listed in the appendix for easy reference or copy-paste functionality. The image requires several changes to be safely rolled out to production.

Log into rc-util1. Access the image you have been assigned.

The pattern for the name is as follows: rmacc-cohort-202601-{node_suffix}-{number}

For example: rmacc-cohort-202601-u1-1-1

The relevant command is: wwctl image shell $imageName

Please make any necessary edits. Roll out.

Services and utilities

Key ideas: Schedulers (Slurm). Software stack. Identity and access management (IAM). Storage. Monitoring. Reporting.

Exercise 3: Slurm testing, validation, and job submission

Nodes are already available in the RC Slurm instance. You are encouraged to check the Slurm config file at /etc/slurm.conf on your node. This exercise will focus on job submission and reporting.

We will begin by establishing accessibility of your node in Slurm.

Submit one interactive job to amilan using sinteractive, asking for 500MB of memory, 1 node, 1 task, and 1 minute of wall time.

Submit a batch job to amilan with the same parameters, printing “hello world”. Schedule it to send you an email at begin and end time. Ensure you request the rmacc_students reservation, or the job will likely not start in reasonable time.

Write and be prepared to discuss any observations around how long each job takes to start.

Next, please visit https://xdmod.rc.colorado.edu/index.php and review the graphs for last month (December 2025). Please answer the following questions:

- Describe a notable pattern in the “Total CPU Hours by Resource (Top 10)” graph. Identify what might be the cause of a gap between 12/17 and 12/22.

- Open the “Usage” tab. Please identify the following information:
    - What user used the most CPU hours in December?
    - What was the maximum number of active users on a given day in December?
    - By core count, what was the size of the largest job run in December?
    - What is the most-utilized partition on Alpine?

Bonus challenge: Write a line in Bash that will print each running job ID, which node(s) it runs on, and what time it is expected to complete. Sort this output from “soonest completion” to “furthest-out completion”. Hint: you can do most of this with in one Slurm command that is available you as a regular (non-administrator) user.

Exercise 4: System Exploration, Metrics, and Reporting

To warm up, please write a short summary of what you have done already up to this point, across the three previous exercises. Please assume that your audience a member of the HPC team who has comparable technical proficiency, but who is not engaged in the work. A couple of sentences for each exercise is sufficient.

4(a): A guided exploration of the CU Boulder Alpine cluster

Imagine you are a new team member who must quickly learn and note your knowledge about the HPC system. Please prepare a technical description of the system answering the following questions about the components of the system.

Answer the following questions, on paper or in a file. Consider this a research project or open-book exam – you can use any resources you’d like, including each other’s knowledge. Your best bet will usually be to start at http://curc.readthedocs.io. If you need to use Alpine, please run exercises from within an sinteractive session using the rmacc_students reservation. (*Hint*: Useful commands for Slurm interaction: sinfo, scontrol, squeue, sacct, sreport)

Compute

- What is the FQDN of the node you are working on?
- Parse and interpret the hostname of the node – what does each part of the name “mean”?
- Access one of the NVIDIA GPU nodes. What CUDA and NVIDIA driver versions are installed?
- What are the default modules loaded into a shell session on Alpine?
- Using an acompile session, compile a simple “Hello world” script written in C. Submit a small job running it. Use default email settings.
- What QOS’s does your user have? What about user crea5307?

Scratch storage (*Hint*: commands interfacing with GPFS will generally begin with mm. IBM developed GPFS, so the documentation is on their site.)

- What’s the name of Alpine’s GPFS filesystem?
- What’s the path to a given user’s Alpine scratch directory?
- How many nodes are in the GPFS cluster?
- What’s the quota set for max number of files on Alpine scratch?
- On how many nodes is Alpine scratch currently mounted?
- What is the purge policy for scratch files?

Network

- What is your node's IP address?
- What is the default route on your login node?

Physical Layer

- Provide a written description in your own words of CU's data center. Include electrical and HVAC notes.

4(b): Communicating to stakeholders

This exercise focuses on describing your success. It is a writing and discussion exercise rather than a command-line or hardware exercise.

The director of your HPC center requires a report from you about Alpine. She requests a one-page writeup, to include the following:

- How many cores do we have? How much storage? How many GPUs? GPU nodes?
- How many active users?
- What has been the average utilization over the past week, month, and year?

The more specific, the better. Please consult an instructor with questions about any details that you believe may be relevant.

Professional success

Key ideas: Success at work. Building connections. Resume. LinkedIn.

- Resumes, cover letters, and related documents
- LinkedIn badge – will send list of names and emails
- Note we can do intros to our own professional networks

Data Center Experience

The data center experience will be conducted by the CU Boulder Data Center Operations (DCOPS) team.

You will have the opportunity to rack and stack hardware, as well as to receive tours of the data center from the Research Computing team.

Safety requirements: You will wear ear protection (if there are more students than sets of provided ear protection, additional earplugs will be provided). Please also watch for tile gaps that could be hazards. Be mindful that this is a relatively small space.

Some items will take place after the event, including: network cabling, frontend/backend networking, baseboard management controller access/configuration, and configuration for PXE Boot.

Return any included packing, power materials, trash, etc. to the Dell box.

Acknowledgements

The instructors thank the Research Computing team for helping make this program a success.

This experience has been funded by the National Science Foundation, grant OAC #2322260, “CC* Regional Computing: Enhancing Computing At Regional Schools in the Rocky Mountain Advanced Computing Consortium (RMACC)”.

Glossary

Alpine: the third supercomputer at CURC; from this is derived the prefix “c3” in node names

CU: “University of Colorado”; letters are intentionally out of usual abbreviation order to reduce ambiguity; alone, the abbreviation refers to the whole campus system not just the Boulder campus

CURC: “CU Boulder Research Computing”

Janus: the first supercomputer at CURC

OIT: “Office of Information Technology”, the IT department at CU

PetaLibrary: CURC storage offering, renting large-scale persistent storage to research groups

RIT: “Research Infrastructure Team”, the on-premises system administration team at CURC

Summit: the second supercomputer at CURC

Appendix

	CPU	Memory	SSD	Base Clock (GHz)	GPUs	Cost
Option 1	64 cores AMD	256 GB	480 GB	3.25	None	$17,803
Option 2	128 cores AMD	512 GB	480 GB	2.45	None	$23,730
Option 3	64 cores AMD	256 GB	1.6 TB	3.25	2 x NVIDIA L40S	$34,370
Option 4	128 cores AMD	512 GB	1.6 TB	2.45	2x NVIDIA H100	$79,978

Reference

My username: _____

My node: _____

My image: _____

A quote for your consideration

Quote No. 237.1
Total$ 36,363.63
Customer #0987654321
Quoted On Jan 31, 2024
Expires on February 29, 2024

Product: SuperNode Blue7005 - No GPU/ 1TB MEM
Unit Price: $36,763.63
Quantity: 1

Subtotal $36,763.63
Shipping $0.00
Non-Taxable Amount $36,763.63
Taxable Amount $0.00
Estimated Tax $0.00
Total: $36,763.63

SuperNode Blue7005 - No GPU/ 1TB MEM
Estimated delivery if purchased today:
October 1, 2025
Contract # H57
Customer Agreement # 76T-110C / CK 1941

Description	SKU	Unit Price	Quantity	Subtotal
SAS/SATA Backplane	KAMTHAYS	-	1	-
SuperNode Blue7005 Server	49VLMF1O	-	1	-
No Trusted Platform Module	5NGDWOXZ	-	1	-
8X 3.5 Chipset SATA / SWRAID S150 with XGMI	L4HAZ0LR	-	1	-
AMD 7543 2.8GHz,32C/64T,256M,225W,3200	937SM6IY	-	1	-
AMD 7543 2.8GHz,32C/64T,256M,225W,3200	NS9JJTVA	-	1	-
Additional Processor Selected	TFYCU9TR	-	1	-
Heatsink for 2 CPU + GPU/FPGA/Full Length card configs	7021T2CK	-	1	
Performance Optimized	0ULL6I7Q	-	1	-
3200MT/s RDIMMs	2RROMXBA	-	1	-
No RAID with Embedded SATA	XB3N77JL	-	1	-
No Controller	XMAUH7A1	-	1	-
Performance BIOS Settings	UZUHPFS1	-	1	-
UEFI BIOS Boot Mode with GPT Partition	QFNOCOO8	-	1	-
High Performance Fan x6	3EP5QZPE	-	1	-
Single, Hot-Plug, Power Supply 2400W Non-Redundant	TUKPJHCY	-	1	-
No Power Cord	NKUDZC38	-	1	-
Riser Config 3 Full Length V2	AP4WN1NT	-	1	-
SuperNode Blue7005 Motherboard, with 2 x 1Gb Onboard LOM, MLK V2	AJFL0PJ1	-	1	-
BMCZone9 Datacenter 15G	T427WOV6	-	1	-

GPU Ready Configuration Cable Install Kit Blue7005	UATP8UWZ	-	1	-
SuperNode 2U Standard Bezel	QS82QFDZ	-	1	-
No Quick Sync	OOKH4FB5	-	1	-
No Operating System	306M22VZ	-	1	-
No Media Required	QPP7YXZS	-	1	-
Sliding Rails	Y6Z2QOTL	-	1	-
Cable Management Arm, 2U	K67MGF5C	-	1	-
Fan Foam, HDD 2U	KAMTHAYS	-	1	-
Assembly BOSS Blank	49VLMF1O	-	1	-
No Systems Documentation	5NGDWOXZ	-	1	-
SuperNode Blue7005 Shipping	L4HAZ0LR	-	1	-
Blue7005 Shipping Material	937SM6IY	-	1	-
SuperNode Blue7005 no CE or with CCC Marking	NS9JJTVA	-	1	-
Warranty 7x24 HW-SW Technical Support and Assistance 5 Years	TFYCU9TR	-	1	-
Warranty Next Business Day On-Site Service After Problem Diagnosis 5 Years	7021T2CK	-	1	-
On-Site Installation Declined	2RROMXBA	-	1	-
64GB RDIMM, 3200MT/s, Dual Rank, 16Gb	XB3N77JL	-	8	-
480GB SSD SATA Read Intensive 6Gbps 512 2.5in Hot-plug AG	XMAUH7A1	-	1	-
Drive, 3.5in HYB CARR, 1 DWPD	UZUHPFS1			

Subtotal $36,763.63

Shipping: $0.00

Estimated Tax: $0.00